Intrusion Detection System using Machine Learning

Introduction to Cybersecurity and Intrusion Detection:

Cybersecurity has become one of the most important challenges in the modern digital world. With the rapid growth of internet usage, cloud computing, IoT devices, digital banking, online education, and smart technologies, cyber attacks are also increasing significantly. Organizations, businesses, universities, banks, and government systems continuously face threats such as malware attacks, phishing, unauthorized access, Denial of Service (DoS), ransomware, and network intrusions.

Traditional security systems often fail to detect modern and evolving cyber threats because many of them rely on fixed rules or predefined signatures. These traditional systems are unable to efficiently identify unknown or advanced attacks, leading to:

- High false alarm rates
- Slow threat detection
- Poor attack classification
- Increased security risks

To overcome these limitations, Machine Learning-based Intrusion Detection Systems (IDS) are becoming highly important in modern cybersecurity environments.

What is an Intrusion Detection System (IDS)?

An Intrusion Detection System (IDS) is a cybersecurity system that monitors network traffic and system activities to identify suspicious behavior, malicious activities, or unauthorized access attempts.

The primary purpose of IDS is:

- To detect cyber attacks
- To monitor abnormal network behavior
- To improve network security
- To protect systems from unauthorized access
- To reduce the risk of data breaches

IDS systems are commonly used in:

- Banking systems
- Government organizations
- Universities
- Corporate networks
- Cloud platforms
- IoT environments
- Data centers

Why Machine Learning is Important in IDS:

Traditional IDS systems mainly rely on rule-based or signature-based methods. Although they can detect known attacks, they often struggle to identify:

- Unknown attacks
- Zero-day attacks
- Complex attack patterns
- Evolving malicious activities

Machine Learning improves IDS systems by:

- Learning patterns from historical network data
- Automatically identifying suspicious behavior
- Improving detection accuracy
- Reducing false positives
- Detecting complex attack patterns more effectively

Machine Learning-based IDS systems are more intelligent, adaptive, and efficient compared to traditional security systems.

Problem Statement:

Modern organizations face increasing cybersecurity threats, while traditional intrusion detection systems often fail to efficiently detect complex and evolving cyber attacks. There is a need for a more intelligent and scalable intrusion detection system capable of accurately identifying malicious network traffic and classifying different types of cyber attacks using advanced machine learning techniques.

Problem Solved by This Project:

This project aims to solve major limitations of traditional intrusion detection systems by using advanced Machine Learning algorithms to automatically analyze network traffic and detect malicious activities.

The proposed system:

- Detects suspicious network traffic
- Identifies different categories of cyber attacks
- Improves intrusion detection accuracy
- Supports intelligent cybersecurity monitoring
- Reduces dependency on traditional rule-based systems
- Provides better attack classification capability

The project uses both Binary Classification and Multi-Class Classification approaches to improve intrusion detection performance and provide more realistic cybersecurity analysis.

Understanding Cyber Attacks in the Dataset:

1. DoS (Denial of Service)

A DoS attack attempts to overload a server or network with excessive requests, making the system unavailable to legitimate users.

Example:
A website becomes inaccessible because attackers continuously send massive traffic requests to the server.

2. Probe Attack

Probe attacks involve scanning networks or systems to identify vulnerabilities and security weaknesses.

Purpose:
Attackers collect information before launching larger attacks.

3. R2L (Remote to Local)

In Remote to Local attacks, an external attacker attempts to gain local access to a target system without authorization.

Example:
An attacker remotely accesses user accounts or files illegally.

4. U2R (User to Root)

In User to Root attacks, a normal user attempts to gain administrator or root privileges.

Purpose:
Attackers try to obtain complete system control.

Dataset Used – NSL-KDD:

This project uses the NSL-KDD dataset, one of the most widely used benchmark datasets for intrusion detection research.

The dataset contains:

- Normal network traffic
- Multiple categories of cyber attacks
- 41 network-related features
- Training and testing datasets

The NSL-KDD dataset is an improved version of the older KDD99 dataset and removes duplicate records to provide better evaluation quality.

Attack Categories in Dataset:

- Normal
- DoS
- Probe
- R2L
- U2R

Dataset Source:
https://www.kaggle.com/datasets/kiranmahesh/nslkdd

Research Baseline:

Our project is based on the research paper:

“Intrusion Detection System Using Ensemble Learning and SHAP Explainability”

The research paper used:

- Random Forest
- XGBoost
- Ensemble Learning
- SHAP Explainability

The paper achieved approximately:

- 98.86% testing accuracy

We selected this paper because:

- It uses the same NSL-KDD dataset
- It focuses on machine learning-based intrusion detection
- It provides strong comparative analysis
- It is recent and closely related to our project objectives

Research Paper Source:
https://www.researchgate.net/publication/403958833_Intrusion_Detection_System_Using_Ensemble_Learning_and_SHAP_Explainability

Project Objective:

The primary objectives of this project are:

- To build a machine learning-based Intrusion Detection System
- To detect malicious network traffic
- To classify different cyber attack categories
- To compare our performance with existing research work
- To improve intrusion detection accuracy using advanced boosting algorithms

Machine Learning Models Used:

1. XGBoost

XGBoost (Extreme Gradient Boosting) is one of the most powerful boosting algorithms used in machine learning and cybersecurity applications.

Key Features:

- High prediction accuracy
- Fast computation
- Efficient handling of large datasets
- Strong classification capability
- Excellent performance on structured data

Reason for Selection:
XGBoost performs extremely well on cybersecurity datasets and is widely used in intrusion detection research.

2. LightGBM

LightGBM (Light Gradient Boosting Machine) is a high-performance boosting algorithm developed by Microsoft.

Key Features:

- Faster training speed
- Lower memory usage
- High scalability
- Efficient gradient boosting
- Strong classification performance

Reason for Selection:
LightGBM provides excellent predictive performance while maintaining fast execution speed and computational efficiency.

Project Workflow:

1. Data Preprocessing

Before training the models, the raw dataset must be cleaned and transformed.

Preprocessing steps include:

- Handling missing values
- Removing duplicate records
- Label Encoding

- Feature Scaling
- Feature Selection

Purpose:
These steps improve data quality and model performance.

2. Exploratory Data Analysis (EDA)

EDA helps understand the dataset and identify important patterns and relationships.

EDA includes:

- Class distribution analysis
- Feature importance analysis
- Correlation heatmaps
- Histograms and visualizations
- Attack category distribution

Purpose:
EDA helps understand how different network features relate to cyber attacks.

3. Binary Classification

In Binary Classification:

- Normal traffic = Normal
- All attacks = Attack

Purpose:

- Direct comparison with the selected research paper
- Benchmark performance evaluation

Models Used:

- XGBoost
- LightGBM

4. Multi-Class Classification

In Multi-Class Classification, the system predicts:

- Normal
- DoS
- Probe
- R2L
- U2R

Purpose:

- More realistic intrusion detection
- Better attack category identification
- Advanced cybersecurity analysis

This approach makes the project more advanced than traditional binary IDS systems.

Models Used:

- XGBoost
- LightGBM

Evaluation Metrics:

Accuracy

Measures overall prediction correctness.

Precision

Measures how many predicted attacks are actually attacks.

Recall

Measures how many real attacks were correctly detected.

F1-Score

Balances precision and recall.

ROC-AUC

Measures the classification capability of the model.

Confusion Matrix

Shows correct and incorrect predictions for each class.

Purpose:
These metrics help evaluate the effectiveness and reliability of the intrusion detection system.

Comparative Analysis:

The final project compares:

- Research paper results
- Binary classification results
- Multi-class classification results
- XGBoost performance
- LightGBM performance

Example Comparison Table:

Model	Type	Accuracy
Research Paper	Binary	98.86%
Our XGBoost	Binary	XX%
Our LightGBM	Binary	XX%
Our Multi-Class Model	Multi-Class	XX%

Purpose:
This comparison helps evaluate whether our models achieve competitive or improved intrusion detection performance compared to existing research work.

Expected Outcome:

The expected outcome of this project is to develop an efficient and accurate Machine Learning-based Intrusion Detection System capable of:

- Detecting malicious network traffic
- Classifying cyber attack categories
- Improving cybersecurity monitoring
- Supporting real-world intrusion detection applications

The project also demonstrates how modern Machine Learning algorithms can significantly improve network security systems and cybersecurity operations.

Conclusion:

This project demonstrates the importance of Machine Learning in modern cybersecurity systems. By combining advanced boosting algorithms such as XGBoost and LightGBM with intrusion detection techniques, the proposed system aims to improve cyber attack detection accuracy and provide better security analysis.

The integration of both Binary Classification and Multi-Class Classification makes the system more effective, realistic, and suitable for modern cybersecurity environments.

This study also highlights how Machine Learning can help organizations build smarter, faster, and more adaptive security systems to protect digital infrastructure from evolving cyber threats.

References / Sources:

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